MILITARY STANDARD

FUZES, NAVY, DESIGN SAFETY CRITERIA FOR

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DEPARTMENT OF THE NAVY

NAVAL AIR SYSTEMS COMMAND

NAVAL ORDNANCE SYSTEMS COMMAND

WASHINGTON, D.C. 20360

Fuzes, Navy, Design Safety Criteria for

MIL-STD-1316 (NAVY)

1. This standard has been approved by the Naval Air Systems Command and the Naval Ordnance Systems Command, and is published to establish design safety criteria for naval fuzes.

2. As of the promulgation date of this document, this standard is a mandatory requirement to be invoked in all specifications or purchase descriptions (and contracts when necessary) in the design of naval fuzes for high explosive warheads used by the Naval Air Systems Command and Naval Ordnance Systems Command.

3. Recommended corrections, additions or deletions should be addressed to the Commanding Officer, Naval Air Engineering Center, Code X, Weapons Engineering Standardization Office (WESO), Philadelphia, Pa. 19112.

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1. SCOPE

1.1 Scope - This standard describes and establishes the fundamental design safety criteria for fuzes, used by the Navy for high explosive warheads (guided missiles, rockets, projectiles, bomblets, bombs), destruct systems, and BW/CW warheads or dispersal systems. It does not apply to fuzes for nuclear warheads.

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2. REFERENCED DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this Standard to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-00339	Tetryl (Trinitrophenylmethyinitramine)
MIL-R-21723	RDX Composition CH-6
MIL-E-81111	PBXN-5

STANDARDS

Military

MIL-STD-1314 (NAVY)	Safety Precautions for Explosive Loaded Items

Publications

WS-4660	Dipam Explosive
WS-5003	HNS, Type I or II, Grade A
NOLC Report 666	Experimental Methods for Determining the Effectiveness of Interruption of a Fuze Explosive Train, 1 Jul 1966

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3. DEFINITIONS

3.1 Definitions - The definitions of MIL-STD-444 apply to ammunition terms used in this standard. For interpretation of this standard, the following specific definitions apply:

a. Assembled fuze: The completed fuze with all component parts put together; the condition of a fuze requiring no added components or parts to prepare it for installation into the round (projectile, rocket, bomb, etc.) in which it is to function. Assembling the fuze is the process of putting the parts and components together.

b. Premature: A malfunctioning in which a munition explodes or functions after launch but before completion of the arming cycle.

c. Dud: A malfunctioning of a munition which results in failure to explode or function, although such was intended.

d. Early burst: A malfunctioning in which a munition explodes or functions after launch and completion of the arming cycle but before the expected time or circumstance.

e. Fuze installation: The act of installing or inserting the assembled fuze into the round (projectile, rocket, bomb, etc.) in which it is to function; sometimes called "fuzing," e.g., fuzing the bomb.

f. Fuze safety system: An electrical, mechanical or electromechanical system consisting of one or more of the following:

(1) Environment sensors,

(2) Launch event sensors,

(3) Command functioned devices,

(4) A logic network, comprising electrical switches and/or mechanical interlocks and timers, utilizing inputs from (1), (2), and (3), above,

which functions to arm the fuze (by closing switches in, and/or removing, interrupters from the firing train) at a safe distance from, or a safe time after, the point at which a normal launch is started, provided the events sensed and commands received have been normal as determined by (1), (2), (3), and (4) above.

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g. Objective: A goal which should be accepted as a requirement unless shown to be not feasible.

h. Design guides: Concepts, logic, background, examples, statements of good practices or rules, or time-proven concepts or features, any of which can be used as reminders or can aid the thought process of the designer in his conscious effort to obtain optimum safety through design.

i. Requirement: A feature or procedure which must be a part of the fuze or fuze development.

j. Safe separation: A space relationship between launcher and weapon, following launch, where an acceptably low probability of injury to personnel or unacceptable damage to the launcher exists if the weapon should function.

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4. GENERAL REQUIREMENTS

4.1 Applicability - The provisions of this standard apply to:

a. Fuze developments to meet future requirements where such developments have not progressed beyond the conceptual phase.

b. Fuze developments in progress when directed, and to the extent directed by the appropriate authority.

c. Fuzes developed by agencies outside the naval establishment intended for Navy use, when directed, and to the extent directed by the appropriate authority.

4.2 Policy - It is the policy of the Naval Air Systems Command (NASC) and the Naval Ordnance Systems Command (NOSC) that safety and arming requirements, objectives, and design guides shall be considered as mandatory fundamental elements of design and in engineering, production and procurement of fuzes. Navy fuzes shall provide safety which is consistent with operational use, operational readiness, and the consequences of a safety failure, and shall require a minimum of special handling over and above prescribed ordnance handling.

4.3 Action - All activities under the direction of the NAVAIR and NAVORD when engaged in the research and development of fuzes for naval weapons within the scope of this Instruction will take necessary action to implement this policy.

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5. DETAIL REQUIREMENTS

5.1 Requirements - The following features, procedures, or controls are requirements and shall be incorporated into the design of naval fuzes or fuzes procured for Navy use:

5.1.1 Explosive train interruption - When the explosive train includes primary explosives (materials more sensitive than lead and booster explosives), an interrupter shall separate the primary explosives from the main charge before arming. The effectiveness of the interrupter shall be determined by techniques described in NOLC Report 666.

5.1.2 Explosive sensitivity - The following explosives only are permitted in a position leading to the initiation of the main charge without interruption when the fuze is in the safe condition:

(a) Tetryl per MIL-T-00339

(b) CH-6 per MIL-R-21723 (Nord)

(c) PBXN-5 per MIL-E-81111 (Wep)

(d) DIPAM, Grade A, per WS-4660B

(e) HNS, Type I or II, Grade A, per WS-5003C.

5.1.3 Electric initiators "in-line" - Electric initiators "in-line" (i.e., not followed by explosive train interruption) shall not be used in fuzes even though explosives employed are those listed in 5.1.2.

5.1.4 Safe-arm indication -

5.1.4.1 One or more of the following options shall be selected or combined in the fuze design:

(a) A positive means of determining safe or armed conditions prior to installation into the round.

(b) A feature which prevents installation of an armed fuze into the round.

(c) A feature which prevents assembling the fuze in the armed condition, and use of arming mechanisms which cannot be accidentally or inadvertently armed during handling.

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Option (a) shall be used in a manner observable after installation into the round; and option (b) shall not be used if, after installation, the fuze could be armed by any method other than by launching of the round. Option (c) alone shall not be used if arming the assembled fuze in tests is a normal procedure prior to firing.

5.1.5 Documentation - The evaluation program used as the basis for safety release shall be documented in both detail and summary forms.

5.1.6 Quality control - Quality control procedure shall be invoked in production to assure that designed safety is maintained. Procurement documentation, provided as a part of the design and release for procurement, shall provide inspection procedures which adequately determine that the products:

(a) Contain the required quality of materials.

(b) Are fabricated to within the proper tolerances on critical dimensions affecting safety.

(c) Are assembled in proper sequence with all critical parameters within limits.

End item inspection will include multiple-personnel inspection including both supplier and government personnel independently inspecting and certifying as to the safe condition of the end item and critical subassembly.

5.1.7 Safety Standard - The fundamental standardized safety precautions necessary for safety loading, assembling, and handling explosive-loaded items are to be found in MIL-STD-1314.

5.2 Objectives - The following characteristics, features, and controls shall be objectives in the design of naval fuzes. Deviations from these objectives may be allowed on submission of proof that compliance is not feasible on the basis of cost, complexity, and compatibility, or is beyond the state-of-art.

5.2.1 Safety failure rate - A safety failure rate not to exceed 1×10^{-6} should be used as a goal in the design of a fuze and, to the extent possible, in its evaluation. This rate is defined as the summation of safety failure in all phases from manufacture up to the intentional initiation of arming.

5.2.2 Stored energy - The fuze design will not incorporate a stored energy mechanism which is used to remove the interrupter of the fuze train.

5.2.3 Safety system - The safety system of fuzes will be composed of at least two series arming mechanisms requiring independent sources of arming energy.

5.2.4 Energy after launch - One of the mechanisms of 5.2.3 will derive its energy from an environmental condition after launching and will be the device which unlocks or arms the interrupted train.

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5.2.5 Quality control - End item inspection will include automatic test equipment when feasible and warranted by producted quantities. The test equipment should be self-checking and designed so that equipment failure rejects rather than accepts the fuze.

5.3 Design Guides - New fuze designs will be guided by, but not restricted to, procedures and features which have proved effective and valuable in fuze developments. The following guidelines will be used where applicable.

5.3.1 Predesign Analysis of Potential Hazards - The purpose of a fuze safety system is to prevent accidental actuation of a weapon during assembly, handling, transportation, and storage, and in operational use until the weapon is a safe distance from the launcher. To assure maximum safety in these phases, the fuze must be designed to provide protection in abnormal environments, conditions and personnel actions which may occur, as well as in the environments, conditions, and actions which are normal. It is less likely that a design safety consideration will be overlooked if as complete a list as possible is made of the abnormal environments, conditions, and personnel actions which may occur in the phases before safe separation and this list is used as a guide to choice of design features. Severe accident situations, such as fire, crushing, and striking a hard surface at high velocity, should be included but judgment and experience must be used to give preference to situations which are more likely to arise. The process of preparing and analyzing this list to express those hazardous conditions in engineering terms (such as environment magnitudes and physical descriptions of personnel actions), which are useful to designers in determining needed characteristics of safety devices, is referred to as potential hazards analysis. The potential hazards analysis is a normal thought process of every designer. Formalizing it in report form is an aid in assuring thoroughness and completeness.

5.3.2 Safety Failure Mode Analysis - A useful method for analyzing the safety of a fuze design is a safety failure mode analysis considering potential accident paths. This analysis is a systematic consideration of the effects on fuze safety of such things as omission of parts, breaking of parts, out of order sequences, and personnel errors. It should consider abnormal as well as normal environments, conditions, and personnel actions. The series of events which can lead to an accident (the accident paths) are examined to determine that adequate precautions to preclude them have been taken in design or through inspection and administrative procedures. As in the potential hazards analysis, severe accident situations should be included and judgment and experience must be used to give more emphasis to situations which are more likely to occur. This analysis differs from the predesign analysis of potential hazards in that it is a study of the fuze safety on the basis of known or anticipated hardware characteristics, whereas the predesign analysis is a study to determine and establish desirable safety characteristics and specifications prior to hardware design.

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5.3.3 Safety Features of Classes of Fuzes - The best features of proven designs should be considered as guidelines for new designs. In this regard the following considerations apply to classes of fuzes. The term "classes of fuzes" used here refers to broad groupings of fuzes used with weapons whose operational release or launch conditions are similar and therefore the forces available for arming during release or launch are similar:

5.3.3.1 Gun-Fired Projectile Fuzes -

(a) Both setback and spin should be used to actuate arming devices in fuzes for projectiles used in rifled guns.

(b) Setback and a bore-rider, or equally effective device, should be used to actuate arming devices in fuzes for projectiles used in smooth bore guns.

(c) Arming should be delayed until the projectile has cleared the muzzle of the gun by a safe separation as determined by experience or effects studies.

5.3.3.2 Aircraft-Carried, Free-Fall Weapon Fuzes -

(a) Operation of at least two arming devices should not occur until one has sensed or been actuated by intentional separation from the aircraft and one has sensed or been actuated by a flight or free-fall environment and these events have occurred in the order given.

(b) After separation from the aircraft, arming should be delayed until a safe separation is reached.

(c) Air vanes, if used, should require a minimum air velocity for operation set as far above non-flight conditions as can reasonably be done without endangering operability.

5.3.3.3 Rocket and Guided Missile Fuzes -

(a) At least one arming mechanism should be operated by acceleration or motor pressure, with acceleration being the preferred choice.

(b) Arming should be delayed until accumulated acceleration-time or pressure-time indicates a safe separation has been reached.

5.3.3.4 Destructor S and A's - Two independent signals should be required for the arming of destructor S and A's.

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5.3.3.5 Bomblet Fuzes -

(a) Actuation of at least one arming device should not occur until it has sensed or been actuated by an operational environment after launch.

(b) Air vanes, if used, should require a minimum air velocity for operation set as far above nonflight conditions existing external to the dispenser as can reasonably be done without endangering operability.

(c) Arming should be delayed until simultaneous actuation of the bomblets could be sustained without disastrous effects to the manned launcher.

(d) An additional lock against arming should be provided by the assembly of the bomblet into the cluster or container. This lock would be released by separation or dispersion of the bomblets.

5.3.3.6 BW and CW Dispersal Controls - Arming of BW and CW dispersal controls should, in general, be guided by the objectives applying to the particular "class" of fuze and weapon system design.

5.3.3.7 Other Fuzes - The arming of fuzes other than those listed in this section should be guided by requirements and objectives applicable to all fuzes and any special provisions directed by the Commander, NAVAIR or Commander, NAVORD

REVIEW ACTIVITY:

Navy - OS

PREPARING ACTIVITY:

Navy - AS

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